

PHYS 3717: Particle Physics

<https://homiller-hep.net/teaching/2026-2/>

Homework 4

Deadline: October 8, 2026 (in class)

Problem 1: Noether Current for Lorentz Transformations

Consider the theory of a free, real scalar field studied in class. The theory is invariant under Lorentz transformations:

$$x^\mu \rightarrow x'^\mu = \Lambda^\mu{}_\nu x^\nu \simeq (\delta^\mu{}_\nu + \omega^\mu{}_\nu) x^\nu = x^\mu + \omega^\mu{}_\nu x^\nu. \quad (1)$$

Show that the conserved Noether current associated with Lorentz invariance can be written

$$j^{\mu\nu\rho} = \frac{1}{2} (T^{\mu\nu} x^\rho - T^{\mu\rho} x^\nu), \quad (2)$$

with $\partial_\mu j^{\mu\nu\rho} = 0$. Here, $T^\mu{}_\nu$ is the conserved stress energy tensor associated with translational invariance in the scalar field theory. Obtain expressions for the conserved charges associated with spatial rotations:

$$Q^{ij} = \int d^3x j^{0ij}. \quad (3)$$

Problem 2: Conserved Current from a Global U(1) Symmetry

Consider the theory of a free *complex* scalar field, described by the Lagrangian

$$\mathcal{L} = (\partial^\mu \varphi^*) (\partial_\mu \varphi) - m^2 \varphi^* \varphi. \quad (4)$$

(a) Show that the Lagrangian is invariant under the global U(1) field transformation,

$$\varphi(x) \mapsto e^{i\alpha} \varphi(x), \quad \alpha = \text{const.} \quad (5)$$

(b) Apply Noether's theorem to derive the following conserved current j^μ corresponding to the global U(1) symmetry:

$$j^\mu(x) = i [(\partial^\mu \varphi^*) \varphi - \varphi^* (\partial^\mu \varphi)]. \quad (6)$$

(Hint: consider the infinitesimal form of the transformation given above. Identify the shift in the configuration of both φ and φ^* and the corresponding shift in the Lagrangian.)

(c) Show explicitly that the current derived above is conserved, $\partial_\mu j^\mu = 0$, by applying the equations of motion for φ and φ^* .

Problem 3: Quantized Field Operators and Mode Expansions

For the following exercises, you may use the Shrödinger picture mode expansions for the quantized scalar field $\phi(x)$ and its conjugate momentum $\pi(x)$:

$$\begin{aligned}\phi(x) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} \left(a_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} + a_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}} \right), \\ \pi(x) &= \int \frac{d^3p}{(2\pi)^3} (-i) \sqrt{\frac{\omega_{\vec{p}}}{2}} \left(a_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} - a_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}} \right),\end{aligned}\tag{7}$$

and the commutation relations for the creation and annihilation operators,

$$\begin{aligned}[a_{\vec{p}}, a_{\vec{q}}^\dagger] &= (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}), \\ [a_{\vec{p}}, a_{\vec{q}}] &= [a_{\vec{p}}^\dagger, a_{\vec{q}}^\dagger] = 0.\end{aligned}\tag{8}$$

(a) Verify the canonical commutation relation,

$$[\phi(\vec{x}), \pi(\vec{y})] = i\delta^{(3)}(\vec{x} - \vec{y}).\tag{9}$$

(b) Show that the Hamiltonian can be written in terms of creation and annihilation operators as

$$H = \int \frac{d^3p}{(2\pi)^3} \omega_{\vec{p}} \left[a_{\vec{p}}^\dagger a_{\vec{p}} + \frac{1}{2} (2\pi)^3 \delta^{(3)}(0) \right].\tag{10}$$

(c) Show that the momentum operator can be written in terms of creation and annihilation operators as

$$\vec{P} = \int \frac{d^3p}{(2\pi)^3} \vec{p} a_{\vec{p}}^\dagger a_{\vec{p}}.\tag{11}$$